



Artificial Intelligence Software, Lecture 4

Data preprocessing



Center for
Cognitive
Modeling



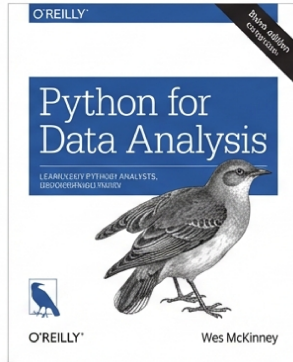
Outline

1. Importance of Data Processing
2. Tabular Thinking
3. Data Hygiene
4. Exploratory Data Analysis
5. Engineering and Encoding

Reference materials



Seminar: Introduction to Data Analysis (Titanic)

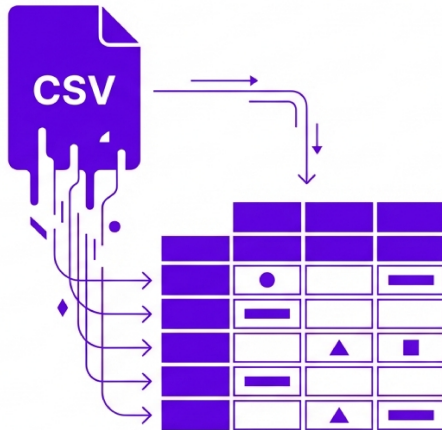


Wes McKinney, Creator of Pandas

Concept: Garbage In, Garbage Out (GIGO)

- Data is rarely ready for analysis in its raw form.
- 80% of an analyst's time is spent on data preparation.
- Models are only as good as the data they consume.

Principle: Quality over Complexity.



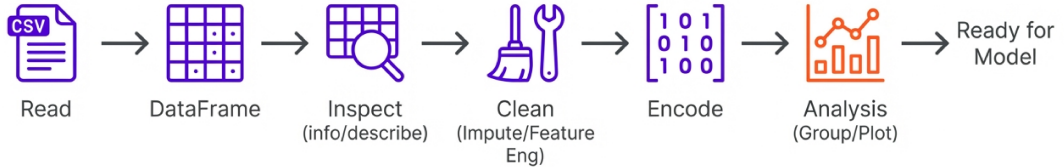
Example: The Titanic Objective

Target Variable
Survived (0 = No, 1 = Yes)
Categorical Features
Sex Embarked (C/Q/S) Pclass (1/2/3)
Numerical Features
Age Fare SibSp Parch

"The goal is to predict survival based on passenger attributes. We face a mix of types: continuous numbers, finite categories, and text strings."



The Analytical Pipeline

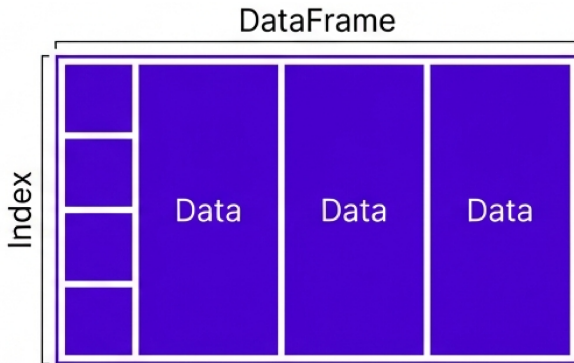
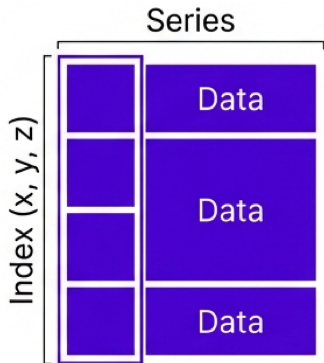




Where to Study

- **Wes McKinney PDF:** Chapter 1 "Preliminaries" (Page 24).
- **Video:** ["Introduction to Data Analysis. Part 1"](#) (Intro section).

Pandas Data Structures





Data Inspection Tools

1

```
data.head()
```

PassengerId	Survived	Pclass	Name	Sex
1	0	3	Braund, Mr. Owen Harris	male
2	1	1	Cumings, Mrs. John Bradley	female
3	1	3	Heikkinen, Miss. Laina	female
4	1	1	Futrelle, Mrs. Jacques Heath	female
5	0	3	Allen, Mr. William Henry	male

2

```
data.info()
```

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

The info() method acts as an X-ray, revealing that "Age" and "Cabin" contain null values (NaN) that require immediate intervention.

3

```
data.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	0.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.38383	2.30864	29.6991	0.52300	0.381594	32.20420
std	257.35384	0.48659	0.83607	14.5264	1.102743	0.80005	48.89342
min	1.000000	0.00000	1.00000	0.42000	0.00000	0.00000	0.000000
25%	223.50000	0.00000	2.00000	20.1250	0.00000	0.00000	7.910400
50%	446.00000	0.00000	3.00000	28.0000	0.00000	0.00000	14.454200
75%	668.50000	1.00000	3.00000	38.0000	1.00000	0.00000	31.00000
max	891.00000	1.00000	3.00000	80.0000	8.00000	6.00000	512.32820



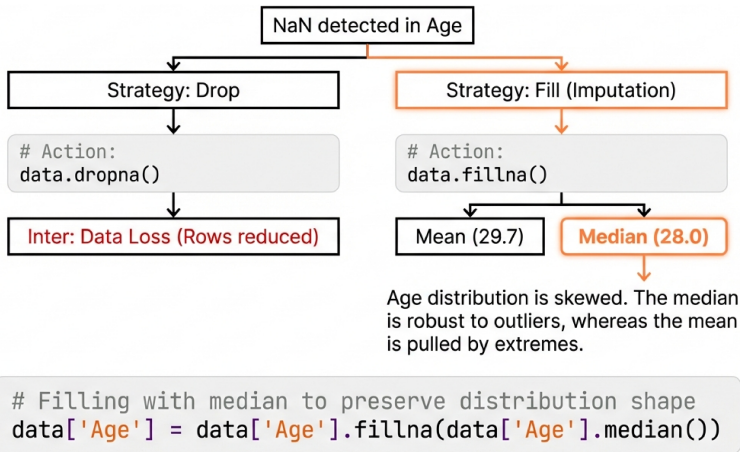
Where to Study

- **Wes McKinney PDF:** Chapter 5 "Getting Started with pandas" (Page 139).
- **Video:** "Introduction to Data Analysis. Part 1" (Intro section).

Strategies for Missing Data

- Missing values (NaN) are a signal, not just empty space.
- Strategy depends on the nature of the data.

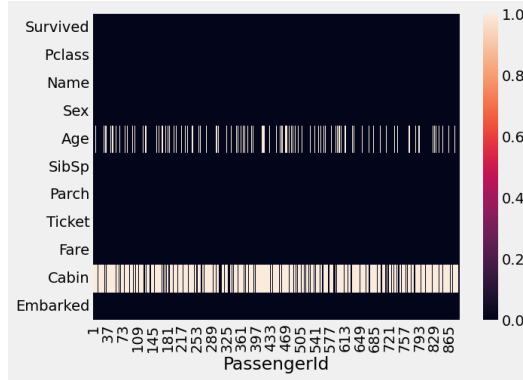
Principle: Handle gaps before they break your model.





Visualizing Data Gaps

Heatmaps (Seaborn) show the pattern of missingness.



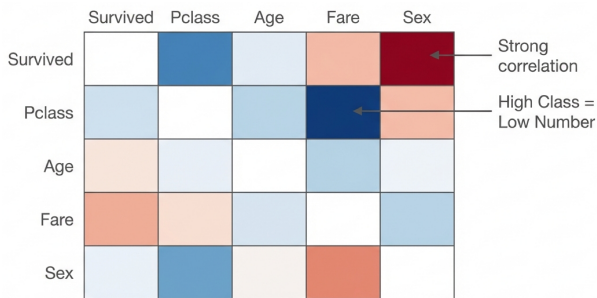


Where to Study

- **Wes McKinney PDF:** Chapter 7 "Data Cleaning and Preparation" (Page 214).
- **Video:** [Introduction to Data Analysis. Part 2](#) (Preprocessing).

Concept: Multivariate Analysis

- Finding linear dependencies between features.
- Avoiding redundancy in models.



Aggregation Logic: Split-Apply-Combine

Initial Table

Sex	Survived
Female	1
Male	0
Female	1
Male	1
Male	1
Male	0
Male	1
Male	1
Male	0
Male	0
Male	1

Split

Female Group

Sex	Survived
Female	1
Male	0
Female	1
Male	1

Male Group

Sex	Survived
Male	0
Male	0
Male	1

Apply

Survived
Σ Avg

Survived
Σ Avg

Combine

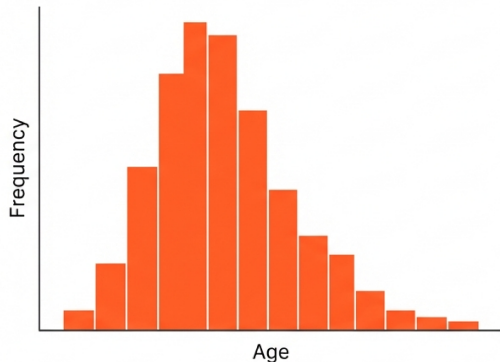
Sex	Mean Survived
Female	0.74
Male	0.18

```
data.groupby('Sex')  
      ['Survived'].mean()
```

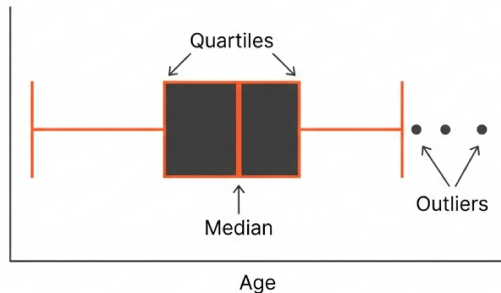
**Women ~74%
Survival Rate
vs Men ~18%.**

Univariate Analysis: Distributions

Age Distribution



Age Spread



Boxplots allow us to instantly spot outliers—passengers with extreme ages or fares—that might skew our models.



Question-Driven Visualization

- "Does ticket class affect survival?"
- Use Bar charts or Pie charts to answer.

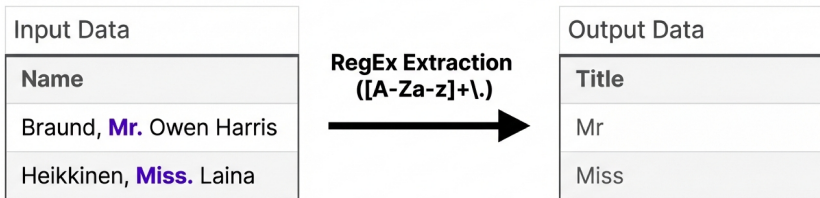


Where to Study

- **Wes McKinney PDF:** Chapter 9 "Plotting and Visualization" (Page 287).
- **Video:** "Introduction to Data Analysis. Part 2" (EDA).

Concept: Feature Engineering

- Creating proxy features for social status and gender.
- Transforming unstructured text into meaningful categories.



```
data['Title'] = data['Name'].str.extract('([A-Za-z]+\.)')
```

Categorical Encoding

- Algorithms cannot multiply "City" by a weight.
- Numbers are the only language models speak.

Label Encoding

Sex Column

Male → 0

Female → 1

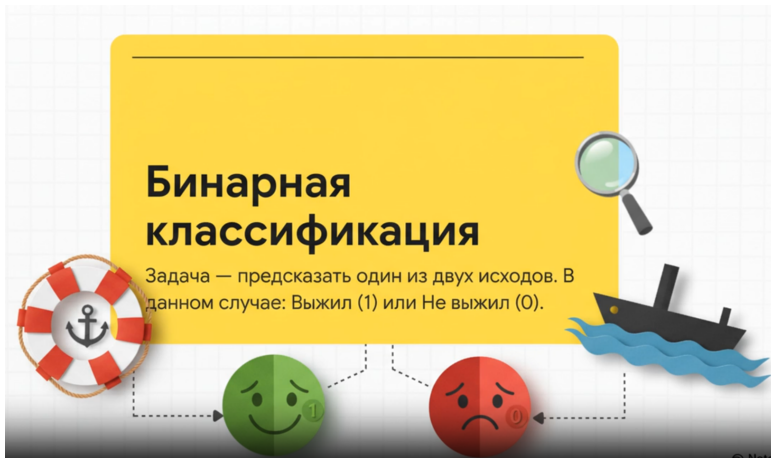
Best for binary or ordinal data.

One-Hot Encoding

Input	Output		
Embarked	Embarked_S	Embarked_C	Embarked_Q
S	1	0	0
C	0	1	0
Q	0	0	1

Recap

Video





Reference materials

- Stepik Deep Learning Course, part 3.2
- «Python и анализ данных: Первичная обработка данных с применением pandas, NumPy и Jupiter» Уэс Маккинни
- Jupiter notebooks in «extra_materials» folder.

Deserve "A" grade!

– Oleg Bulichev

✉ bulichev.ov@mipt.ru

📍 @Lupasic

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